Xinjie Liu

CONTACT INFORMATION

5.104 Peter O'Donnell Jr. Building (POB),
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RESEARCH INTERESTS

My research focuses on developing theoretical foundations and practical algorithms for decision-making and control in autonomous systems operating under uncertainty. I am particularly interested in (i) efficient and scalable reinforcement learning through principled integration of data from multiple sources, and (ii) intelligent and safe interaction among multi-agent systems, such as robots operating in shared environments.

EDUCATION

The University of Texas at Austin, Austin, TX
Ph.D. in Electrical and Computer Engineering — Aug 2023 - present
Advisors: Ufuk Topcu and David Fridovich-Keil

Delft University of Technology, Delft, Netherlands
M.Sc. in Robotics, GPA: 9.04/10.0 (top 1.6%, *cum laude*) — Sep 2021 - Jul 2023
Advisor: Javier Alonso-Mora

Tongji University, Shanghai, China
B.Eng. in Automotive Engineering, GPA: 88.9/100 — Sep 2016 - Jul 2021

Graz University of Technology, Graz, Austria
Exchange Program — Jan 2020 - Jun 2020

PUBLICATIONS

[8] **X. Liu***, C. Neary*, K. Gupta, W. A. Suttle, C. Ellis, U. Topcu, and D. Fridovich-Keil. A Multi-Fidelity Control Variate Approach for Policy Gradient Estimation, Transactions on Machine Learning Research (TMLR), 2025.

[7] Y. Jain*, **X. Liu***, L. Peters*, D. Fridovich-Keil, and U. Topcu. Bayesian Inverse Games with High-Dimensional Multi-Modal Observations, under review at The International Journal of Robotics Research (IJRR), 2025.

[6] **X. Liu**, J. Li, F. Fotiadis, M. O. Karabag, J. Milzman, D. Fridovich-Keil, and U. Topcu. Approximate Feedback Nash Equilibria with Sparse Inter-Agent Dependencies, under review at IEEE Transactions on Automatic Control (T-AC), 2025.

[5] K. Gupta, **X. Liu**, U. Topcu, and D. Fridovich-Keil. Second-Order Algorithms for Finding Local Nash Equilibria in Zero-Sum Games, under review, 2025.

[4] C. Armstrong*, R. Park*, **X. Liu**, K. Gupta, and D. Fridovich-Keil. Inferring Foresightedness in Dynamic Noncooperative Games, IEEE Robotics and Automation Letters (RA-L), 2025.

[3] **X. Liu**, J. Li, F. Fotiadis, M. O. Karabag, J. Milzman, D. Fridovich-Keil, and U. Topcu. Policies with Sparse Inter-Agent Dependencies in Dynamic Games: A Dynamic Programming Approach, Autonomous Agents and Multi-Agent Systems (AAMAS), Extended Abstract, 2025.

[2] **X. Liu***, L. Peters*, J. Alonso-Mora, U. Topcu, and D. Fridovich-Keil. Auto-Encoding Bayesian Inverse Games, Algorithmic Foundations of Robotics XVI (WAFR), 2024.

[1] **X. Liu***, L. Peters*, and J. Alonso-Mora. Learning to Play Trajectory Games against Opponents with Unknown Objectives, IEEE Robotics and Automation Letters (RA-L), 2023.

*indicates equal contribution.

Technical Reports & Theses

[3] **X. Liu**. On Game-Theoretic Planning with Unknown Opponents' Objectives, master's thesis, Delft University of Technology, 2023. (Mentor: Javier Alonso-Mora)

[2] **X. Liu** and V. Atanassov. Safe Model Predictive Control Approach for Non-Holonomic Mobile Robots, Delft University of Technology, 2022. (Mentor: Sergio Grammatico)

[1] **X. Liu**, R. M. Rodríguez, P. Féry, and Y. Zhang. Planning Algorithm for a Quadrotor Drone, Delft University of Technology, 2022. (Mentor: Javier Alonso-Mora)

Honors and Awards

Cockrell School of Engineering Fellowship, UT Austin 2023-present

Travel Grant, Texas Robotics 2025

Travel Grant, Workshop on the Algorithmic Foundations of Robotics (WAFR) 2024

Graduation *cum laude*, TU Delft (top 1.6% of MSc program) 2023

First Prize, **Franka Emika Robot Hackathon Challenge**, European Robotics Forum 2022

National Scholarship for Undergraduate Exchange Program (1% of department) 2020

Annual Excellent Student Award, Tongji University (top 5% of department) 2020

First Prize Academic Scholarship, Tongji University (top 3% of department) 2019

Winning Prize for Outstanding Innovative Projects, Tongji University 2019
Project: Autonomous Flight of UAVs Based on UWB Localization

Third Prize Academic Scholarship, Tongji University (top 20% of department) 2018

Scholarship for Social Activities, Tongji University 2017, 2019

Projects

Web | GitHub

High-Precision Robot Assembly Challenge Jun 2022 - Jul 2022
Team Member, Delft University of Technology (1/6)
- Developed a point cloud-based perception module and interactive imitation learning technique to tackle high-precision robot assembly tasks on a randomly positioned board
- Winner of the Franka Emika Challenge at the European Robotics Forum Hackathon

Autonomous Robotic Field Coverage Apr 2022 - Jun 2022
Multidisciplinary Project with Lely, Main Contributor (1/5)
- Designed a robotic field coverage system with obstacle avoidance and battery management on a Husky robot, including modules for perception (top-view camera), task planning, path and trajectory planning, and control

MPC Approach for Non-Holonomic Mobile Robots in Dynamic Scenarios Jan 2022 - Apr 2022
Model Predictive Control Course Project, Main Contributor (1/2)
- Proposed an MPC approach with linearized constraints in velocity space for dynamic obstacle avoidance of mobile robots, proved Lyapunov stability for time-varying regulation problems

RNN-Based Event Camera Data Processing Jan 2022 - Apr 2022
Deep Learning Course Project, Main Contributor (1/3)
- Reproduced partial results of "High Speed and High Dynamic Range Video with an Event Camera" using a different dataset, reconstructing intensity images from event data with RNNs

Deep Reinforcement Learning Library Jan 2022 - Apr 2022
Deep Reinforcement Learning Course Project
- Implemented a library including main policy gradient (REINFORCE, Actor-Critic, PPO, DDPG, TD3) and value function (DQN, Double DQN, n-step target, semi-gradients) based methods

Autonomous Quadrotor Delivery System Oct 2021 - Jan 2022
Planning & Decision Making Course Project, Main Contributor (1/4)
- Developed a navigation pipeline for a quadrotor, including global path planning (RRT*), minimum-snap trajectory generation, and nonlinear MPC for tracking and obstacle avoidance

Multisensor Perception for Autonomous Vehicles Oct 2021 - Jan 2022
Machine Perception Course Project
- Developed a self-driving car perception module, including pedestrian detection (CNN, SVM) and vehicle ego-motion compensation using iterative closest point (ICP) with LiDAR data

Learning Autonomous Car Racing from Pixels Sep 2021 - Nov 2021
Machine Learning for Robotics Course Project, Main Contributor (1/2)
- Developed a pipeline for learning driving policies from pixels, including data augmentation, feature extraction, dimensionality reduction, and control (random forest, SVM, neural networks)

Robot Dynamics & Control Sep 2021 - Nov 2021
Robot Dynamics & Control Course Project
- Implemented force and impedance controllers with singularity-robust and task-priority control for a 2-DOF robot arm; implemented PD and nonlinear geometric controllers for quadrotor tracking tasks

Interactive Imitation Learning in Robotics Oct 2020 - Jul 2021
Bachelor's Thesis Project
- Investigated interactive imitation learning methods using human feedback for various simulated robotic tasks, demonstrating improved task performance and data efficiency compared to reinforcement learning baselines

SOFTWARE

MultiFidelityPolicyGradients: Mixes limited target-domain data with abundant low-fidelity data to construct a variance-reduced policy gradient estimator for improved RL sample efficiency

AutoEncodingBayesianInverseGames.jl: Embeds a differentiable Nash game solver into a generative model for amortized inference of continuous distributions in multi-agent interactions

NumericalOptimizationAlgorithms: Numerical optimization algorithms for unconstrained and constrained problems accelerated using JAX

DifferentiableAdaptiveGames.jl: A differentiable Nash equilibrium solver and an inverse game solver for equilibrium-constrained bilevel optimization problems

SafeMPCObstacleAvoidance: MPC planner for non-holonomic robots with linearized dynamic obstacle avoidance constraints in the velocity space

QuadrotorPlanningControl: Navigation pipeline for a quadrotor robot with a global path planner, minimum-snap trajectory generator and MPC controller for obstacle avoidance

MEDIA

Franka Emika Challenge winner at the European Robotics Forum Franka Robotics, 2022

INVITED TALKS

Center for Autonomy Seminar, UT Austin 2025
A Multi-Fidelity Control Variate Approach for Policy Gradient Estimation

Center for Autonomy Seminar, UT Austin 2024
Inferring Multimodal Uncertainty for Game-Theoretic Interactions
Computing Sparse Policies for Noncooperative Dynamic Games

Texas Robotics Seminar, UT Austin 2023
On Game-Theoretic Planning with Unknown Opponents' Objectives

Lockheed Martin Corporation 2023
On Game-Theoretic Planning with Unknown Opponents' Objectives

Princeton University 2023
Learning to Play Trajectory Games Against Opponents with Unknown Objectives

Academic Mentorship

Ryan Park, Computer Science Honors & Aerospace Engineering, UT Austin (next: PhD student at University of California, Berkeley)

Outreach

STEM Muse Mentorship Program, Austin — Sep 2024 - present
Mentoring to provide individualized professional development and career support for women

Girlstart Texas, Austin — Sep 2023 - present
Organizing innovative workshops to increase girls' interest and engagement in STEM fields

Professional Activities

Reviewer for Journals

- IEEE Transactions on Robotics (T-RO)
- IEEE Transactions on Automatic Control (T-AC)
- IEEE Robotics and Automation Letters (RA-L)
- IEEE Transactions on Intelligent Transportation Systems (T-ITS)

Reviewer for Conferences

- International Conference on Learning Representations (ICLR)
- IEEE International Conference on Robotics and Automation (ICRA)
- International Symposium on Robotics Research (ISRR)
- IEEE Conference on Decision and Control (CDC)
- IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)

Graduate-Level Courses

Analysis: Real Analysis, Functional Analysis (basics)
Probability: Probability and Stochastic Processes
Optimization: Convex Optimization, Nonlinear Programming
Machine Learning: Deep Reinforcement Learning, Deep Learning, Machine Learning for Robotics
Control & Game Theory: Game-Theoretic Modeling of Multi-Agent Systems, Model Predictive Control
Robotics: Human-Robot Interaction, Machine Perception, Planning & Decision Making, Robot Dynamics & Control, Robot Software Practicals
Software: Object-Oriented Scientific Programming with C++

Skills

Programming Languages: Python, Julia, C++, Matlab
Version Control: Git
Machine Learning & Numerical Computation: PyTorch, JAX, Zygote.jl, Flux.jl, ChainRules.jl, TensorFlow, Keras
Optimization: PATH Solver, JuMP.jl, YALMIP, CasADi, CVX, FORCES Pro, IPOPT, OSQP
Other: Robot Operating System (ROS), Linux, LaTeX
Languages: Mandarin (native), English (C1, IELTS 7.5), German (B2, DSH 2)

Management Experience

Business Consulting Intern, IQVIA Shanghai — Jul 2020 - Aug 2020
Marketing project for rare disease drugs: conducted policy and business analysis to develop strategies for national negotiations and local market entry of orphan drugs